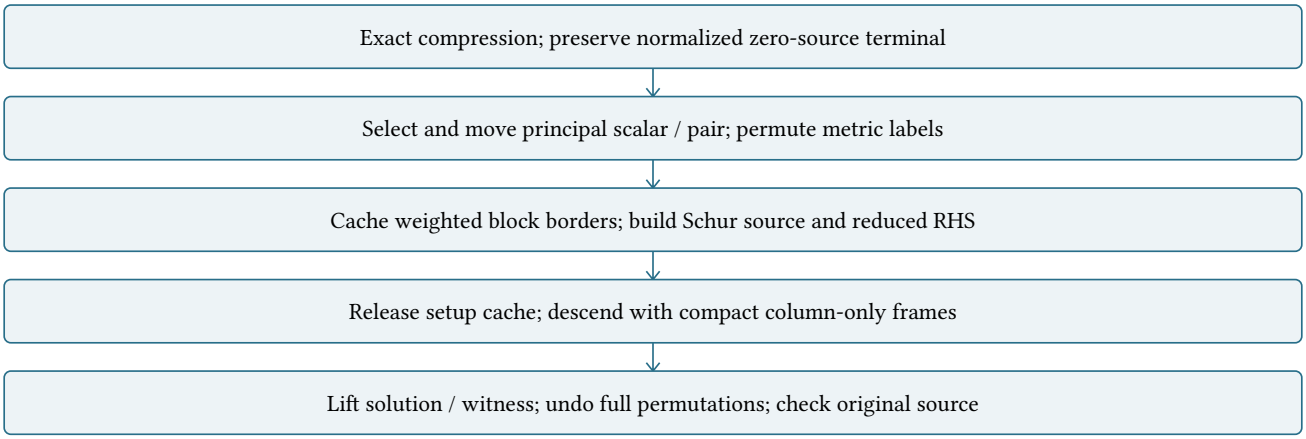
Complete weighted-symmetry solve / witness route

2026-11-02 · Derived construction / static checks · Numerical trials: zero

Domain. Square distinct-wrap companion source over a caller-certified prime $p>3$, with nonzero diagonal metric D and established $DA=A^{T}D$. The public weighted Gram compiler supplies this invariant. Inverse-metric and local reciprocal checks alone do not establish global symmetry.



Complete control flow. Every nonzero source has an invertible principal scalar or antidiagonal pair under the weighted-symmetry assumption. Simultaneous permutations and principal Schur reduction preserve the inherited metric. Full-size pivots use explicit small inverses; zero sources return zero solution or normalized coordinate witness. No matrix solve oracle is supplied.

Block cache and lifts. Leading rows are reconstructed from selected columns using $A_{ji}=d_j^{-1}d_iA_{ij}$. Leading-supported requests use cached matrix-vector arithmetic. For $A=(E,F;C,H)$, solution lift uses $E^{-1}(b_1-Fz)$, with $Fz=D_1^{-1}C^{T}D_2z$; witness lift uses $-E^{-T}C^{T}v$. Only C, RHS head, small inverse, shared root metric and coordinate labels survive in each frame. Full permutation inversion is explicit. Cache and successful frames release their stored arrays.

Derived schedule. Let cut $k\in\{1,2\}$, $q=n-k>0$ and w be transported width. Search/transport/compression, root checks and common Schur work are separate.

Phase	Source calls	Field products	Field additions
Row setup + validation	0	$n+2nk$	0
Supported setup / RHS	0	$nk(2w+5)$	$nk(2w+5)$
Outer Schur boundaries	2	kernel work	kernel work
Primal lift incl. inverse	0	$kq+q+2k$	$kq+k$
Dual lift incl. inverse	0	$kq+k$	$kq+k$

The general uncached block route has $2w+8$ source calls over these phases; this route has two. Scalar and pair pivot sequences differ, so this is a same-block schedule comparison, not a measured full-run improvement. Zero lift inputs still pay the displayed arithmetic. Native and field products remain separate units; transport width growth is still charged.

Storage. Each frame has $kq+2k$ field slots and two length-n integer tuples. The shared root metric costs $2N$ field slots once. Setup temporarily holds $2nk$ border entries alongside captured C. Certificates, root/current sources, metric gathering, meter blocks and trace history are additional memory; no total or measured peak-memory claim follows from these slot counts.

Checks and validation. Exact AST equality for the preserved zero terminal and final root solution/witness checks; all Python files parse and dependency hashes are recorded. No numerical, mutation, heap or performance trial. Full upstream cohorts must cover pair/scalar/zero terminals, both lift directions, composed permutations, metric/border corruption, counters and heap aliases. The execution-cost gate remains in force; feature annihilation stays separate.

Sources / next. driver.py, block_cache.py, block_frame.py, check_preparation.py and DERIVATION.md. Dumas–Pernet2018 (arXiv:1802.10453) and Bostan et al.2017 (DOI 10.1137/16M1062855) are established context, without inherited fast bounds. Next derive paired transport reuse and provenance-bound metric validation.